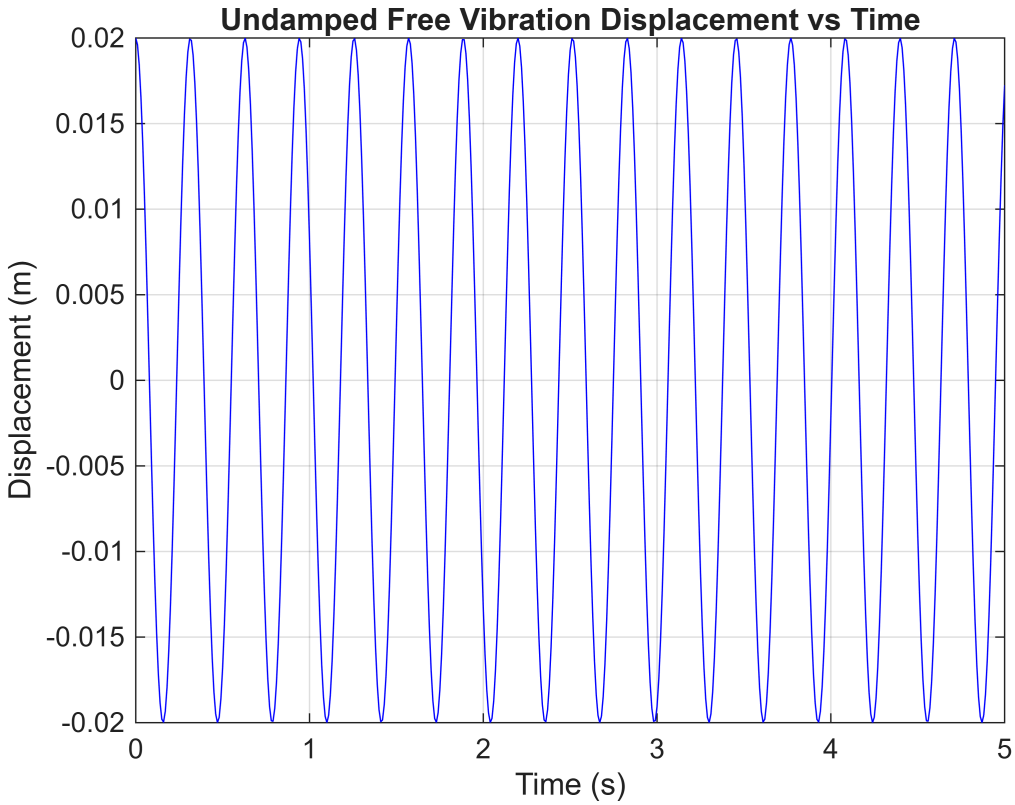
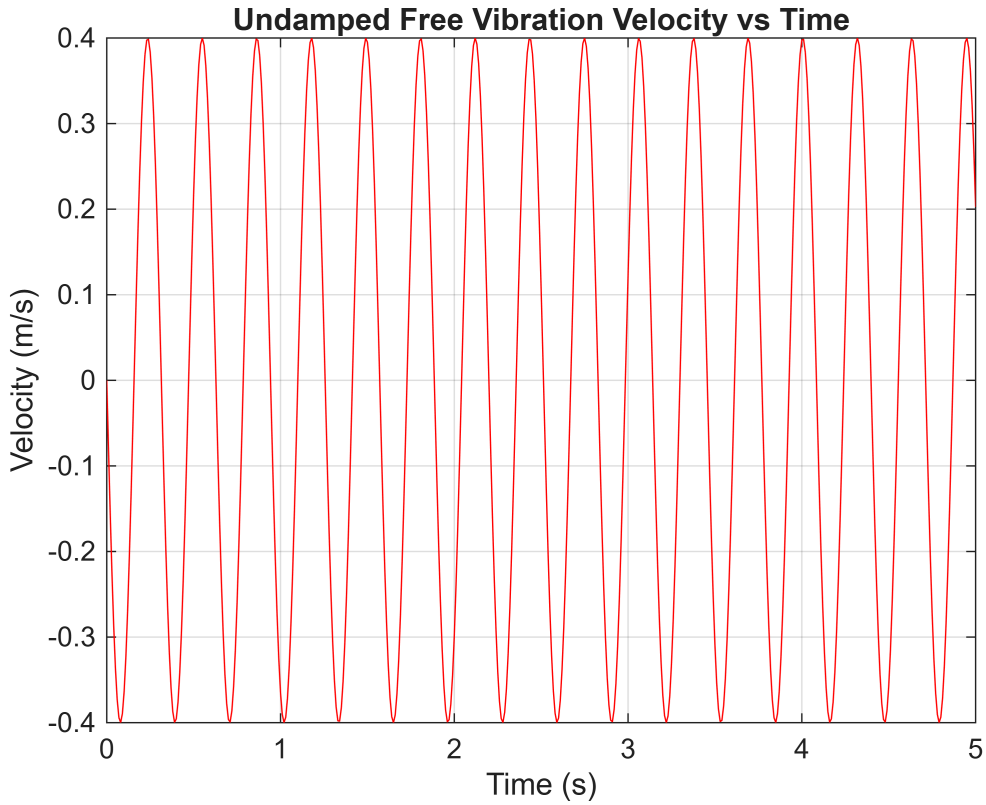
```
%Assignment 1                           :Undamped free vibration
%Name                                   :Ziphozethu Magqazolo
%Student number                         :16622685
%Module                                 :SME 3701


%%Section 1.1                           :Define given variables
m     = 5;                              %mass in kg
k     = 2000;                           %stiffness in N/m]
x0    = 0.02;                           %displacement in m

%Determine natural angular frequency wn
wn    = sqrt(k / m);                    %natural angular frequency in rad/s
%Calculate                              :wn = sqrt (2000 / 5 ) = 20 rad/s


%%Section 1.2                            :Displacement calculations
%Create time vector
t     = (0:0.01:5);                     %Time intervals from 0 to 5 seconds

%Compute displacement using below formula
x     = x0 * cos(wn * t);               % Displacement as a function of time
%Calculate @t = 0                        :x = 0.02 * cos (20 * 0) = 0.02m
%Calculate @t = 0.01                     :x = 0.02 * cos (20 * 0.01) = 0.02m
%Calculate @t = 5                        :x = 0.02 * cos (20 * 5) = 0.02m
%Displacement is constant


%%Section 1.3                           :Plot displacement vs time
plot(t, x, 'b');
xlabel('Time (s)');
ylabel('Displacement (m)');
title('Undamped Free Vibration Displacement vs Time');
grid on;
```




```
%%Section 1.4                              :Velocity calculations
%Compute velocity
v       = -x0 * wn * sin(wn * t);         % Velocity as a function of time
%Calculate @t = 0                         :v = -0.02 * 20 * sin (20 * 0) = 0
%Calculate @t = 0.01                      :v = -0.02 * 20 * sin (20 * 0.01) =
-0.08 m/s
%Calculate @t = 5                         :v = -0.02 * 20 * sin (20 * 5) = 0.2
m/s
%Calculate vmax = wx                      :vmax = 20 * 0.02 = 0.4 m/s

%Plot velocity of the system
plot(t, v, 'r');
xlabel('Time (s)');
ylabel('Velocity (m/s)');
title('Undamped Free Vibration Velocity vs Time');
grid on;
```




```
%%Section 1.5                                :Analysis and comments

%Nature of motion                            :Simple harmonic motion, mass
oscillates between 0.2 and -0.2

%Effect of zero damping                      :No damping, no energy loss.
Displacement remains constant

%Relationship between velocity and displacement
%Displacement use cos, velocity use sin. They are 90 degrees out of phase.
%When velocity is at peak, displacement is zero. When velocity is zero,
displacement is at peak
```

https://github.com/Ziphozethu-Magqazolo/16622685/blob/main/16622685.pdf